

Color Models – Lecture Notes

■ 1. ACHROMATIC COLOR

- Black, white, grey: $r = g = b$
- No hue, saturation; only brightness (light intensity)

■ 2. COLOR Gamut & Chromaticity

- Color gamut: Range of reproducible colors in a device.
- Chromaticity diagram shows visible color space and hue relationships.

■ 3. COLOR SYSTEMS & MODELS

Color Model: Defines how colors are represented numerically.

Common Models:

- RGB (Red, Green, Blue) – Additive model
- CMY/CMYK (Cyan, Magenta, Yellow, Black) – Subtractive model
- HSV (Hue, Saturation, Value)
- HLS (Hue, Lightness, Saturation)
- YIQ / YUV – Used in TV and video signals

■ 4. RGB MODEL

- Used in screens and displays (Additive)
- $(0,0,0)$ = Black, $(1,1,1)$ = White
- $r = g = b$ = grayscale

■ 5. CMY & CMYK MODEL

- Used in printers (Subtractive)
- $C = 1 - R$, $M = 1 - G$, $Y = 1 - B$
- CMYK adds true black (K) because CMY $\neq$ perfect black in practice

■ 6. HSV MODEL

- Hue (H): Actual color angle on a 360° wheel
- Saturation (S): Purity; 0 = gray, 1 = vivid
- Value (V): Brightness; 0 = black, 1 = full color

■ Conversion RGB $\rightarrow$ HSV:

1. Normalize RGB (if needed)
2. Compute max, min, diff
3. Hue based on max component
4. Saturation = diff / max
5. Value = max

■ HSV $\rightarrow$ RGB:

1. Compute $C = V \times S$
2. Compute $X = C \times (1 - |(H/60 \bmod 2) - 1|)$